EXCEL

Excel for Data Analysts

The 15 functions that cover real work, the 2 debugging checklists interviewers love, and 15 practice questions.

The 15 functions

Function	Use it for	Example
XLOOKUP	Any lookup, left or right	`=XLOOKUP(A2,Emp[ID],Emp[Name],"Not found")`
INDEX + MATCH	Lookups in older Excel	`=INDEX(C:C,MATCH(A2,A:A,0))`
SUMIFS	Sum with conditions	`=SUMIFS(Sales,Region,"North",Month,E1)`
COUNTIFS	Count with conditions	`=COUNTIFS(Status,"Open",Owner,G2)`
IFS / IF	Buckets and flags	`=IFS(B2>=90,"A",B2>=75,"B",TRUE,"C")`
IFERROR	Clean error output	`=IFERROR(B2/C2,0)`
TRIM / CLEAN	Remove hidden spaces	`=TRIM(CLEAN(A2))`
TEXT	Format dates and numbers	`=TEXT(A2,"mmm-yy")`
LEFT / MID / RIGHT	Extract codes	`=MID(A2,4,3)`
TEXTSPLIT	Split by delimiter	`=TEXTSPLIT(A2,",")`
UNIQUE	Distinct list	`=UNIQUE(Region)`
FILTER	Dynamic subsets	`=FILTER(Data,Data[Sales]>50000)`
SORT / SORTBY	Ranked lists	`=SORTBY(Name,Sales,-1)`
EOMONTH	Month-end reporting	`=EOMONTH(A2,0)`
LET	Readable long formulas	`=LET(t,SUM(B:B),B2/t)`

Checklist: VLOOKUP returns #N/A for half the data

1. **Hidden spaces.** Wrap the lookup value in TRIM.
2. **Numbers stored as text** on one side. Convert with VALUE or Text to Columns.
3. **Range not locked.** Use $ or a Table so the range does not shift when copied down.
4. Bonus: the 4th argument is missing, so it is doing an approximate match. Use FALSE.

Checklist: SUM shows 0 but data is there

1. Numbers stored as text (left aligned, green triangle).
2. Calculation set to Manual. Formulas tab, Calculation Options, Automatic.
3. Circular reference somewhere in the sheet.

Pivot tables in 5 steps

1. Convert data to a Table (Ctrl+T).
2. Insert, PivotTable.

3. Category to Rows, metric to Values.
4. Date to Columns, then group by Month.
5. Value Field Settings, Show Values As, % of Grand Total.

15 practice questions

1. Lookup a value that sits to the left of the key column.
2. Total sales for North region in March only.
3. Flag orders above the average order value.
4. Extract the domain from an email ID.
5. Count unique customers.
6. Rank salespeople within each region.
7. Convert 20261105 into a real date.
8. Find duplicates across two sheets.
9. Month-end date for every transaction.
10. Create Low / Medium / High buckets.
11. Top 5 products by revenue, auto-updating.
12. Percentage contribution of each region.
13. Days between order and delivery, excluding weekends.
14. Combine first and last name with a space.
15. Build a dropdown that filters a table.

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